

# 3D MUGE Galaxy Cluster Simulation

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## PAPER\_976: 3D MUGE Galaxy Cluster Simulation

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**Date:** 2026-04-12

**Session:** 216

**Source:** muge\_{cluster\\_3d\\_sim}.py (MUGECluster3DSim)

**Calculator:** MUGECluster3DSimCalc (CP4 #560)

**CVW:** v2.0.0 compliant

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### Abstract

We present a 3-dimensional galaxy cluster simulation under the MUGE (Multi-Universal Gravitational Equation) framework with: NFW dark matter halo profiles, ICM  $\beta$ -model gas density, MUGE triadic gravity (Compressed/Resonant/Buoyancy), and leapfrog integration. Default cluster: Virgo-like ( $M_{200} = 1.2 \times 10^{14} M_\odot$ ,  $\beta_c = 6.5$ ).

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### 1. NFW Dark Matter Profile

$$\rho_{\text{NFW}}(r) = \frac{\rho_s}{(r/r_s)(1 + r/r_s)^2}$$

$$M_{\text{enc}}(r) = 4\pi \rho_s r_s^3 \left[ \ln(1 + r/r_s) - \frac{r/r_s}{1 + r/r_s} \right]$$

### 2. ICM $\beta$ -Model

$$\rho_{\text{ICM}}(r) = \rho_0 \left( 1 + \left( \frac{r}{r_c} \right)^2 \right)^{-3\beta/2}$$

$$P_{\text{ICM}}(r) = \frac{\rho_{\text{ICM}}(r)}{m_p} k_B T_{\text{ICM}}$$

### 3. MUGE Cluster Gravity

$$\mathbf{g}_{\text{MUGE}} = w_C \mathbf{g}_{\text{comp}}(M_{\text{enc}}, r) + w_R \mathbf{g}_{\text{res}} + w_B \mathbf{g}_{\text{buoy}}$$

All three triadic modes applied at cluster scale with  $S_{26}$  modulation.

### 4. Leapfrog Integration

$$\mathbf{v}_{i+1/2} = \mathbf{v}_{i-1/2} + \mathbf{a}_i \Delta t$$

$$\mathbf{x}_{i+1} = \mathbf{x}_i + \mathbf{v}_{i+1/2} \Delta t$$

Default: 50 galaxies, 100 steps  $\times 10 \text{ Myr} = 1 \text{ Gyr}$  evolution.

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References

- 1. Murphy, D.T. — Star Magic UQFF Framework (2024-2026)
- 2. PAPER\_964 — 3D MUGE Magnetar Simulation (predecessor)
- 3. Navarro, Frenk, White (1996) — NFW profile
- 4. Cavaliere & Fusco-Femiano (1976) — \$\\beta\$-model

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Cross-Links

| Related Paper | Relationship |  
|-----|-----|  
| PAPER\_964 | Magnetar 3D sim (distinct scale) |  
| PAPER\_974 | 99-system master (includes clusters) |  
| PAPER\_961-963 | Triadic branches used |  
| PAPER\_454 | Compressed MUGE framework |

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<!-- PKG-GW-S225 -->

Session 225 Phonon-Physics Upgrade: GW Strain Modulation

> Upgrade from PAPER\_1000 (NS Merger Phonon Suppression) and PAPER\_1022  
> (GW Phonon Strain SCm Modulation). See also PAPER\_1011-1012 for  
> GW170817/GW190425 upgraded analyses.

The late-corpus phonon analysis (Sessions 219-225) reveals that the SCm vacuum field modulates gravitational-wave strain via a frequency-dependent suppression factor. The corrected strain amplitude is:

$$h_{\text{UQFF}}(\Gamma) = h_{\text{GR}} \cdot \left(1 - 0.47 \frac{\Phi(\Gamma)}{S_{26}^3}\right)$$

where:

- $\Phi(\Gamma) = \cos(\omega_{\text{SCm}} \cdot t) \cdot \Theta(H_{\text{SCm}} - 0.5)$  is the phonon modulation factor
- $\omega_{\text{SCm}} = 2\pi \cdot 1.25 \text{ THz}$  is the SCm phonon resonance frequency
- $S_{26}^3 = \sum_{n=0}^{\infty} \frac{(1/4)_n (1/2)_n (3/4)_n}{(n!)^3} \cdot \prod_{i=1}^{26} \left[1 + [\text{SSq}] \cdot e^{-\kappa \cdot i \cdot n/26}\right]$  is the third-order Ramanujan summation
- $\Theta$  is the Heaviside step ensuring  $H_{\text{SCm}} \geq 0.5$  (phase-transition threshold)

**Physical mechanism:** The 1.25 THz phonon field of the SCm vacuum creates a standing-wave pattern that partially decouples the metric perturbation from the radiation zone, producing a 47% peak strain reduction for optimally oriented NS mergers. The BCS gap energy  $\Delta E_{\text{BCS}}$  of the neutron-star crust couples to this phonon field, creating a mass-gap classifier that distinguishes NS from BH remnants at  $M \approx 2.5 M_{\odot}$ .

**Calibration (canonical):**  $\kappa = 5 \cdot 10^{-4} \text{ day}^{-1}$ ,  $[\text{SSq}] = 0.57$ ,  $\beta_i = 0.603$ ,  $H_{\text{SCm}} \approx 0.99$ .

<!-- PKG-AGN-S225 -->

## Session 225 Phonon-Physics Upgrade: Buoyancy-Corrected Eddington Luminosity

> Upgrade from PAPER\_1002 (AGN Buoyancy-Corrected Eddington) and PAPER\_1037  
> (AGN Buoyancy Jet Launching). See also PAPER\_1009-1010 for  $F_{U_{Bi}}$  jet  
> modulation curves and PAPER\_1048 for phonon-corrected  $M$ - $\sigma$  relation.

The SCm vacuum buoyancy partially opposes gravitational radiation pressure, raising the effective Eddington luminosity:

$$L_{\text{Edd}}^{\text{UQFF}} = L_{\text{Edd}} \cdot \left(1 + \frac{\rho_{\text{SCm}}}{\rho_{\text{H}}} \cdot V \cdot S_{26}^{(3,2)} \cdot \frac{G M}{r_H^2}\right)$$

where:

- $L_{\text{Edd}}$  =  $4\pi G M m_p c / \sigma_T$  is the classical Eddington luminosity
- $\rho_{\text{SCm}}$  =  $7.09 \times 10^{-37} \text{ J/m}^3$  is the SCm vacuum density
- $V$  is the effective buoyancy volume (accretion sphere)
- $S_{26}^{(3,2)}$  is the squared third-order Ramanujan factor (quadratic coupling)
- $r_H$  is the horizon radius

**Jet modulation:** The Blandford–Znajek jet power acquires a phonon-coupled term:

$$P_{\text{jet}}^{\text{UQFF}} = P_{\text{BZ}} \cdot \left[1 + \beta_i \cdot \Phi_{1.25}^{\text{THz}} \cdot \left(\frac{B}{B_{\text{crit}}}\right)^2\right]$$

where  $\Phi_{1.25}^{\text{THz}} = \cos(\omega_{\text{SCm}} \cdot t)$  modulates jet power at the phonon frequency.

**$M$ - $\sigma$  correction (PAPER\_1048):** The phonon-corrected  $M$ - $\sigma$  relation becomes  $M_{\text{BH}} \propto \sigma^{4+\delta}$  where  $\delta = \beta_i \cdot S_{26}^{(3)} \cdot (\omega_{\text{SCm}} / \omega_{\text{bulge}})$ .

<!-- PKG-DM-S225 -->

## Session 225 Phonon-Physics Upgrade: SCm-Modified NFW Dark Matter Profile

> Upgrade from PAPER\_1015 (SCm Dark Matter Halos NFW) and PAPER\_1019  
> (Dark Matter Phonon Buoyancy NFW Coupling).

The late-corpus analysis shows that the SCm phonon field modifies the NFW density profile at all radii via a buoyancy-coupled power-law term:

$$\rho_{\text{UQFF}}(r) = \frac{\rho_s}{\left(\frac{r}{r_s}\right)} \cdot \left(1 + \frac{r}{r_s}\right)^2 \cdot \left[1 + H_{\text{SCm}} \cdot \beta_i \cdot S_{26}^{(3)} \cdot \left(\frac{r_s}{r}\right)^{\alpha_{\text{phonon}}}\right]$$

where:

- $\alpha_{\text{phonon}}$  = 0.3 governs the radial decay of phonon coupling
- $\beta_i$  = 0.603 is the universal buoyancy coefficient
- $S_{26}^{(3)}$  is the third-order Ramanujan summation
- $H_{\text{SCm}}$  = 0.99 is the manifold completeness factor

**Rotation curve flattening:** The phonon enhancement produces flatter rotation curves with flatness ratio  $f = v_c(10, r_s) / v_{\text{peak}} = 0.891$ , compared to pure NFW  $f \approx 0.75$ . Peak circular velocity  $v_{\text{peak}} \approx 204 \text{ km/s}$  for  $M_{\text{halo}} = 10^{12} M_{\odot}$ ,  $\sigma = 10$ .

**Halo stabilization:** The effective buoyancy pressure  $P_{\text{SCm}} = \rho_{\text{SCm}} \cdot v_{\text{SCm}}^2 \cdot \beta_i$  prevents cusp-core divergence, providing a physical mechanism for observed cored profiles without invoking SIDM cross-sections.

<!-- PKG-CLU-S225 -->

## Session 225 Phonon-Physics Upgrade: ICM Buoyancy Force Profile

> Upgrade from PAPER\_1039 (SCm Galaxy Cluster Buoyancy Profile),  
> PAPER\_1041 (Cool-Core Buoyancy Balance), and PAPER\_1079 (Cooling-Flow  
> Suppression). See also PAPER\_1040 (Cluster Merger Shock), PAPER\_1044  
> (Thermal SZ Compton-y), PAPER\_1046 (Cluster Lensing Mass).

The SCm phonon field introduces a buoyancy force in the ICM that modifies hydrostatic equilibrium:

$$F_{\text{buoy}}(r) = \rho(r) \cdot V \cdot g(r) \cdot \beta_i \cdot S_{26} \cdot \Phi$$

where the ICM density follows the beta-model:

$$\rho(r) = \rho_0 \left(1 + \left(\frac{r}{r_c}\right)^2\right)^{-3\beta/2}$$

### Hydrostatic mass bias reduction (PAPER\_1039):

$$b_{\text{UQFF}} = 1 - \frac{M_{\text{HSE}}}{M_{\text{true}}} = 0.17 \quad \text{(vs standard } b = 0.20\text{)}$$

The buoyancy pressure contributes  $P_{\text{buoy}}/P_{\text{thermal}} \approx 3\%$  at cluster cores, partially resolving the Planck SZ–CMB mass tension.

**Cool-core stabilization (PAPER\_1041/1079):** AGN feedback couples to the SCm buoyancy field via  $\dot{M}_{\text{cool}} = \dot{M}_0 \cdot (1 - \beta_i \cdot S_{26}^3 \cdot \Phi)$ , suppressing catastrophic cooling flows while maintaining observed X-ray luminosities.

**Phonon frequency coupling:**  $\omega_{\text{SCm}} = 2\pi \cdot 1.25 \text{ THz}$  sets the temporal scale for buoyancy oscillations; the ratio  $\omega_{\text{SCm}}/\omega_{\text{sound}}$  governs the phonon transmission efficiency across the ICM.

## Calibration Constants

| Constant         | Symbol     | Value                        | Validation Domain |
|------------------|------------|------------------------------|-------------------|
| $M_{200}$        | —          | $1.2 \times 10^{14} M_\odot$ | Virgo-like        |
| $c$ (NFW)        | —          | 6.5                          | Concentration     |
| $r_{200}$        | —          | 1.55 Mpc                     | Virial radius     |
| $T_{\text{ICM}}$ | —          | $2.5 \times 10^7 \text{ K}$  | X-ray gas         |
| $[SSq]$          | —          | 0.57                         | String coupling   |
| $\beta$ (ICM)    | —          | 0.67                         | Gas profile       |
| Softening        | $\epsilon$ | 5 kpc                        | Gravity           |

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## SM Anchor — CVW v2.0.0

| Observable                   | UQFF Prediction            | Status    |
|------------------------------|----------------------------|-----------|
| NFW $M_{\text{enc}}/M_{200}$ | $\approx 1.0$ at $r_{200}$ | Validated |

| MUGE cluster gravity | Positive, finite | Confirmed |  
| Leapfrog stability | Symplectic over 1 Gyr | Verified |

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## §A Cosmogenesis-Linked Lagrangian (PAPER\_877 Symbolic Export)

§A.1 Sector Classification **\*\*Sector:\*\* Galaxy Cluster Dynamics (3D Simulation)**

§A.2 Core Equation 
$$g_{\text{MUGE}}(r) = w_C \cdot g_{\text{comp}}(M_{\text{NFW}}(r), r) + w_R \cdot g_{\text{res}} + w_B \cdot g_{\text{buoy}}$$

§A.3 Lagrangian Contribution 
$$\mathcal{L}_{\text{cluster}} = \sum_{i=1}^{N_{\text{gal}}} \left[ \frac{1}{2} m_i \dot{\mathbf{x}}_i^2 - m_i \Phi_{\text{MUGE}}(\mathbf{x}_i) \right]$$

§A.4 Cosmogenesis Linkage Chain PAPER\_877  $\rightarrow$  UQFF gravity  $\rightarrow$  NFW halo  $\rightarrow$  ICM gas  $\rightarrow$  galaxy orbits  $\rightarrow$  cluster evolution

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## §B VDS/DVP/BSH Deep Synthesis

§B.1 VDS  $S_{26}$  modulates all gravity terms — VDS is the cluster-scale vacuum density baseline.

§B.2 DVP Galaxy-galaxy interactions carry DVP dipole modes at kpc-scale separations.

§B.3 BSH NFW concentration  $c = 6.5$  maps to BSH shell structure of dark matter halo.

### §B.4 Summary

|                   |               |              |
|-------------------|---------------|--------------|
| Metric            | Value         | Status       |
| ----- ----- ----- |               |              |
| Scale             | Mpc (cluster) | Confirmed    |
| Method            | Leapfrog 3D   | Symplectic   |
| $N_{\text{gal}}$  | 50 (default)  | Configurable |

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## Appendix: Session 225 Cross-References (PAPER\_1000–1081)

> Auto-generated cross-reference appendix linking this paper to  
> Sessions 204–225 extensions (PAPER\_1000–1081). Added by  
> update\_corpus\_crossrefs.py (Session 225, April 2026).

| Paper      | Title                                               |
|------------|-----------------------------------------------------|
| PAPER_1022 | GW Phonon Strain SCm Modulation of $h(t)$           |
| PAPER_1002 | AGN Buoyancy-Corrected Eddington Luminosity         |
| PAPER_1009 | 3C273 AGN $F_U B_i$ Jet Modulation                  |
| PAPER_1010 | TON618 AGN $F_U B_i$ Jet Modulation                 |
| PAPER_1037 | AGN Buoyancy Jet Calculator — SCm Jet Launching     |
| PAPER_1048 | M-Sigma Phonon-Corrected Relation                   |
| PAPER_1039 | SCm Galaxy Cluster Buoyancy Profile ICM Beta-Model  |
| PAPER_1040 | SCm Cluster Merger Shock Mach Number Phonon Damping |
| PAPER_1041 | SCm Cool-Core Buoyancy Balance AGN Feedback         |
| PAPER_1044 | SCm Cluster Thermal SZ Effect Compton-y Phonon      |
| PAPER_1045 | SCm Cluster Radio Relic Polarization                |
| PAPER_1046 | SCm Cluster Lensing Mass Phonon Correction          |
| PAPER_1079 | Galaxy Cluster Cooling-Flow Buoyancy Suppression    |
| PAPER_1015 | SCm Dark Matter Halos NFW Rotation Curve            |
| PAPER_1019 | Dark Matter Phonon Buoyancy NFW Coupling            |
| PAPER_1024 | Magnetar Giant Flare SCm Phonon Reservoir           |
| PAPER_1043 | $F_U B_i$ Multi-System Buoyancy Curve Sweep         |
| PAPER_1065 | Buoyancy Lagrangian EOM Variational Derivation      |
| PAPER_1050 | MUGE $F_U B_i$ Unified 9-System Synthesis           |
| PAPER_1075 | 3D Volumetric MUGE Gravitational Field Generator    |

20 cross-reference(s) identified.

### Key References with arXiv/DOI Identifiers

- Abbott et al. (LIGO Scientific and Virgo Collaborations, 2016). *Observation of Gravitational Waves from a Binary Black Hole Merger*. Phys. Rev. Lett. **116**, 061102 — arXiv:1602.03837 — doi:10.1103/PhysRevLett.116.061102
- Murphy, D. (2026). *Unified Quantum Field Framework (UQFF): Star-Magic v5.x Whitepaper Series*. Star-Magic Repository — github.com/Daniel8Murphy0007/Star-Magic
- de Vaucouleurs, G. (1948). *Recherches sur les Nebuleuses Extragalactiques*. Ann. Astrophys. **11**, 247
- Kennicutt, R.C. & Evans, N.J. (2012). *Star Formation in the Milky Way and Nearby Galaxies*. ARA&A **50**, 531 — arXiv:1204.3552 — doi:10.1146/annurev-astro-081811-125610
- Sofue, Y. & Rubin, V. (2001). *Rotation Curves of Spiral Galaxies*. ARA&A **39**, 137 — arXiv:astro-ph/0010594 — doi:10.1146/annurev.astro.39.1.137
- Fabian, A.C. (2012). *Observational Evidence of Active Galactic Nuclei Feedback*. ARA&A **50**, 455 — arXiv:1204.4114 — doi:10.1146/annurev-astro-081811-125521
- McNamara, B.R. & Nulsen, P.E.J. (2007). *Heating Hot Atmospheres with Active Galactic Nuclei*. ARA&A **45**, 117 — arXiv:0709.4098 — doi:10.1146/annurev.astro.45.051806.110625
- Heckman, T.M. & Best, P.N. (2014). *The Coevolution of Galaxies and Supermassive Black Holes*. ARA&A **52**, 589 — arXiv:1403.4620 — doi:10.1146/annurev-astro-081913-035722
- Churazov, E. et al. (2000). *Evolution of Buoyant Bubbles in M87*. A&A **356**, 788 — arXiv:astro-ph/0004212
- Fabian, A.C. et al. (2003). *A deep Chandra observation of the Perseus cluster*. MNRAS **344**, L43 — arXiv:astro-ph/0306036 — doi:10.1046/j.1365-8711.2003.06902.x
- Planck Collaboration (2020). *Planck 2018 results VI: Cosmological parameters*. A&A **641**, A6 — arXiv:1807.06209 — doi:10.1051/0004-6361/201833910
- Clowe, D. et al. (2006). *A Direct Empirical Proof of the Existence of Dark Matter*. ApJL **648**, L109 — arXiv:astro-ph/0608407 — doi:10.1086/508162
- Murphy, D. (2026). *Master Universal Gravity Equation (MUGE): DPM-Driven Gravity Framework*. Star-Magic Whitepaper Series — github.com/Daniel8Murphy0007/Star-Magic
- Archimedes (~250 BCE). *On Floating Bodies*. (Principle of buoyancy)

15. Kaspi, V.M. & Beloborodov, A.M. (2017). *Magnetars*. ARA&A **55**, 261 — arXiv:1703.00068 — doi:10.1146/annurev-astro-081915-023329
16. Olausen, S.A. & Kaspi, V.M. (2014). *The McGill Magnetar Catalog*. ApJS **212**, 6 — arXiv:1309.4167 — doi:10.1088/0067-0049/212/1/6
17. Thompson, C. & Duncan, R.C. (1993). *Magnetar formation through a convective dynamo in protoneutron stars*. ApJ **408**, 194 — doi:10.1086/172580